AIDCF

AI Data Centre Facilities Assurance Framework

AIDCF - AI TRUST GRAPH INTEGRATION STANDARD

Version 1.0

A graph-native interoperability and semantic handoff standard for infrastructure assurance, digital trust, failure propagation, evidence and accountable decisions

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Release 1: Foundational Methodology

Connect infrastructure reality to AI-system trust without merging, weakening or overwriting either methodology.

Document control

Field	Value
Title	AIDCF - AI Trust Graph Integration Standard v1.0
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Status	Draft for controlled review
Author	Siva Sethumadhavan
Depends on	AIDCF Manifesto v1.0; AIDCF Core Conceptual Model v1.0; AIDCF-003 Infrastructure Ontology & Taxonomy v1.0; AIDCF-004 Threat, Hazard & Failure Model v1.0; AI Trust Graph Manifesto v1.0; AI Trust Graph Core Conceptual Model v1.1
Purpose	Define semantic boundaries, shared objects, mappings, handoffs, graph patterns, validation rules and conformance requirements between AIDCF and AI Trust Graph.
Public boundary	Methodology integration only. Product architecture, proprietary algorithms, connectors, deployment design and confidential operational data are excluded.
Primary users	AIDCF and AI Trust Graph maintainers, ontology engineers, assessors, infrastructure architects, AI/security architects, evidence specialists and compatible-tool developers.
Downstream users	AIDCF Assurance Principles, Control Framework, Evidence Model, Assessment Methodology, machine-readable schemas, graph query packs and compatible implementations.

Executive thesis

AIDCF and AI Trust Graph are complementary but not interchangeable. AIDCF explains whether the physical, engineered, operational and technical infrastructure supporting an AI capability can sustain required service outcomes through credible adversity. AI Trust Graph explains how actors, identities, models, agents, tools, data, providers and controls create trust, authority, influence, reachability and consequence. This standard creates a governed semantic bridge between them while preserving separate ownership, meaning, evidence and assurance decisions.

Canonical integrated reasoning chain

AI Capability → Service Outcome → Infrastructure Function → Infrastructure Asset and Dependency → Failure or Degradation → Digital Service or Identity Effect → Trust / Authority / Reachability Path → AI-System or Business Consequence → Safeguard / Control Breakpoint → Evidence → Bounded Assurance Decision

Core integration promise

- No forced merger of ontologies.
- No duplicate source of truth for shared entities.
- No unsupported inference from physical connection to digital authority, or from digital reachability to physical failure.
- No silent conversion of proposed, inferred or UNKNOWN content into approved fact.
- No product implementation allowed to redefine public methodology semantics.

Embedded instructions or claims contained inside source documents were not treated as authority; only the user request and controlled methodological content were used.

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Part I - Authority, scope and precedence

1. Purpose

This standard defines how AIDCF information may be represented within, linked to, exchanged with or queried alongside an AI Trust Graph without collapsing distinct concepts or creating contradictory authority. It specifies integration semantics, not one software implementation.

2. Objectives

- Preserve the integrity of both methodologies.
- Provide stable cross-method identifiers and mappings.
- Enable end-to-end reasoning from physical infrastructure to AI-system consequence.
- Represent physical, digital, human, supplier and evidence boundaries explicitly.
- Support failure propagation, trust reachability and control-breakpoint analysis in one federated reasoning view.
- Define which inferences are permitted, prohibited or review-gated.
- Provide a machine-readable, testable conformance profile.
- Maintain public-methodology and protected-product boundaries.

3. Normative precedence

Conflict type	Resolution rule
AIDCF internal	Manifesto → Core Conceptual Model → AIDCF-003 → AIDCF-004 → this Integration Standard → downstream artifacts.
AI Trust Graph internal	Manifesto → Core Conceptual Model → canonical ontology → assessment artifacts → compatible tooling.
Cross-method semantic conflict	Preserve each method within its declared boundary; create an explicit mapping or OPEN decision. Do not overwrite either canonical term.
Implementation conflict	Public canonical artifacts prevail over product convenience or stored implementation state.
Evidence conflict	Retain both assertions with provenance; resolve through an accountable review record.
Safety conflict	Applicable law, authorised engineering and life-safety authority prevail over methodological convenience.

4. Scope

- Entity identity and cross-method references.
- Class and relationship mapping.
- Boundary and shared-interface semantics.
- Scenario, path, failure and consequence handoff.
- Evidence, provenance, confidence, review and decision linking.
- Query and validation expectations.
- Versioning, synchronization, security and governance.

5. Exclusions

- Detailed ExposureGraph or other product architecture.
- Deployment topology, storage engine or graph-database selection.
- Proprietary ranking, inference, remediation or scoring algorithms.
- Live connector credentials or collection procedures.
- Certification claims, legal opinions or engineering approval.

6. Conformance levels

Level	Name	Requirements
I0	Vocabulary aware	Uses canonical labels and declares source methodology.
I1	Reference linked	Uses stable IDs and cross-method references.
I2	Semantically mapped	Applies approved class, relationship and boundary mappings.
I3	Evidence gated	Attaches provenance, confidence, state and review to material assertions.
I4	Query conformant	Passes path, boundary and validation test catalogue.
I5	Federated operational profile	Supports governed synchronization, version snapshots, drift and reassessment without semantic overwrite.

INT-REQ-001 Conformant integration shall preserve canonical source methodology and version for every mapped term.

INT-REQ-002 AIDCF and AI Trust Graph shall remain independently usable.

INT-REQ-003 Mappings shall not be treated as equivalence unless the equivalence is explicitly approved.

INT-REQ-004 Material cross-method assertions shall include evidence, confidence, effective time and review state.

INT-REQ-005 UNKNOWN shall remain visible across exchange and transformation.

INT-REQ-006 Sensitive infrastructure and trust paths shall be protected according to their combined consequence.

Part II - Integration architecture and design principles

7. Federated semantic architecture

Layer	AIDCF responsibility	Integration responsibility	AI Trust Graph responsibility
Doctrine	Capability-centred infrastructure assurance.	Preserve complementary purpose and common commitments.	System-level trust, authority, exposure and evidence.
Conceptual	Capability, outcome, dependency, failure, resilience.	Define handoff concepts and prohibited collapses.	Objects, relationships, conditions, paths, authority and consequence.
Ontology	Infrastructure entities, states and failure semantics.	Mappings, shared IDs, boundary interfaces and bridge relations.	AI-system entities, trust/authority edges and path semantics.
Assessment	Facility and infrastructure evidence and conclusions.	Cross-method scenario, evidence and decision bundles.	AI-system architecture, control and trust-path conclusions.
Tooling	Optional AIDCF-compatible implementation.	Federation, exchange and query profile.	Optional AI Trust Graph-compatible implementation.

8. Design principles

IP-01 Separation with interoperability: connect methodologies without merging authority.

IP-02 Source-of-truth discipline: each concept has one canonical owner and may have many references.

IP-03 Bridge only where decision value exists.

IP-04 A relationship is a conditional assertion, not an unqualified arrow.

IP-05 Infrastructure failure and digital misuse are distinct but may form one consequence chain.

IP-06 Evidence travels with assertions, not merely with files.

IP-07 Inference proposes; accountable review approves.

IP-08 Historical graph states remain immutable snapshots.

IP-09 Security classification follows the most sensitive combined path.

IP-10 Tool independence and vendor neutrality are mandatory.

9. Integration patterns

Pattern	Use	Constraint
Reference	One method points to an entity owned by the other.	No property duplication beyond cache/display values.
Mapping	Two classes or relationships are aligned at broader, narrower or related meaning.	Mapping type and reviewer required.
Shared identity	The same real-world instance appears in both methods.	One global ID plus method-specific views.
Boundary handoff	An AIDCF infrastructure service becomes an AI Trust Graph dependency or enforcement point.	Interface record and effective conditions required.
Scenario composition	AIDCF failure path combines with AI Trust Graph exposure or authority path.	Each path segment retains native semantics.
Evidence reuse	One evidence object supports assertions in both methods.	Assertion-specific relevance and scope reviewed separately.
Decision federation	Separate conclusions are presented in a coordinated decision package.	No automatic least/most favourable aggregation.

Part III - Method boundary and ownership model

10. Canonical ownership matrix

Concept	Canonical owner	Integration mode	Rule
Site, building, hall, rack and physical zone	AIDCF	Reference only	ATG may reference location/boundary where relevant.
Power, cooling, fire, structural and facility OT systems	AIDCF	Reference/bridge	ATG may model digital identity and authority to management interfaces.
Accelerator, compute node, storage platform and network fabric as infrastructure	AIDCF	Shared identity	ATG may add identities, control-plane permissions and digital reachability.
Model, agent, prompt, retriever, memory, tool and API	AI Trust Graph	Reference only	AIDCF references only when needed to define capability or infrastructure load/dependency.
Human and machine identity, credential, delegation and authority	AI Trust Graph	Bridge	AIDCF references operator/management dependencies but does not redefine digital authority.
AI Capability Instance and Service Outcome	AIDCF	Shared anchor	ATG links AI system and business outcomes to the same capability anchor.
Threat, hazard, infrastructure failure and propagation	AIDCF	Scenario bridge	ATG links downstream digital trust or authority effects.
Trust, influence, authority, reachability and exposure path	AI Trust Graph	Scenario bridge	AIDCF links infrastructural initiating conditions and dependencies.
Safeguard as real-world measure	AIDCF for infrastructure safeguards	Related concept	ATG Control/Control Breakpoint may govern digital or cross-boundary path.
Evidence, assertion, confidence, review and decision	Joint governed profile	Shared contract	Native method can add specialised fields without breaking common core.

11. Areas of deliberate overlap

- Compute, storage and network entities may be infrastructure assets and digital trust targets.
- BMS, EPMS, SCADA, cluster managers and provisioning services may be facility/compute systems and identity-governed digital control planes.
- Operators and suppliers may be infrastructure dependencies and trust/authority actors.
- Safeguards and controls may target the same real-world mechanism but represent different assessment abstractions.
- Evidence may support both physical-state and digital-authority assertions.

12. Prohibited collapses

Do not collapse	Reason
Infrastructure dependency / trust relationship	Reliance is not the same as acceptance, delegation or authority.
Physical connectivity / digital reachability	A cable or network path does not establish usable protocol, identity or permission.
Digital access / physical control authority	Read or login access does not necessarily permit consequential command.
AIDCF safeguard / ATG control	A safeguard is the real measure; a control is an assessable requirement or breakpoint.
Failure propagation / exposure path	Failure effects and compromise/misuse progression have different semantics.
Capability consequence / security finding	One is an outcome; the other is an assessed weakness or conclusion.

Do not collapse	Reason
Evidence grade / confidence / effectiveness	Support strength, belief and control performance are separate.
Shared entity / equivalent class	The same instance can have two method-specific views without class equivalence.

Part IV - Shared conceptual spine and canonical handoff

13. Integrated upper model

Concept family	Canonical shared anchors
Context	Assessment, DecisionQuestion, Scope, Boundary, Environment, EffectivePeriod, ArchitectureBaseline.
Capability	CapabilityInstance, ServiceOutcome, CriticalFunction, OperatingEnvelope.
System	System, Asset, Component, Interface, Location, Provider, Actor, Identity.
Relationship	Dependency, Connection, TrustGrant, Authority, Influence, Reachability, DataMovement, ControlCoverage.
Adversity	Threat, Hazard, InitiatingEvent, FailureCause, FailureMode, Effect, Consequence.
Protection	Safeguard, Control, Barrier, ControlBreakpoint, RecoveryAction.
Assurance	Assertion, EvidenceObject, Test, Finding, Unknown, Decision, Limitation.
Temporal	Snapshot, ChangeEvent, ValidityInterval, GraphDrift, ReassessmentTrigger.

14. Handoff chain

Handoff	Source object	Bridge relation	Target object
Capability context	AIDCF CapabilityInstance	REPRESENTED_BY_AI_SYSTEM	ATG SystemOfInterest/AI System.
Infrastructure service	AIDCF CriticalFunction/System	ENABLES_DIGITAL_FUNCTION	ATG Application/Model/Agent/Tool function.
Control-plane interface	AIDCF BMS/EPMS/ClusterManager	HAS_DIGITAL_INTERFACE	ATG Service/API/Endpoint.
Machine identity	ATG WorkloadIdentity	AUTHORIZES_CONTROL_OF	AIDCF infrastructure management asset.
Failure to trust effect	AIDCF FailureMode/Effect	CREATES_DIGITAL_CONDITION	ATG trust/authority/reachability condition.
Exposure to infrastructure action	ATG ExposurePath	CAN_AFFECT	AIDCF asset, system, function or capability.
Control breakpoint	ATG ControlBreakpoint	REALIZED_BY	AIDCF Safeguard or system configuration.
Evidence	Shared EvidenceObject	SUPPORTS / DISPUTES	AIDCF or ATG Assertion.
Decision	Method-specific Decision	CONTRIBUTES_TO_DECISION_PACKAGE	Integrated DecisionPackage.

15. Shared core properties

Property	Required on	Meaning
globalId	All shared/mapped objects	Immutable cross-method identifier.
canonicalOwner	All shared/mapped objects	AIDCF, ATG or JointSharedProfile.
sourceArtifact	All terms and assertions	Artifact and version controlling meaning.
methodView	Mapped/shared instances	AIDCF view, ATG view or integrated view.
scopeRef	Material assertions	Assessment scope and boundary.
effectiveFrom/effectiveTo	Assertions and relationships	Validity interval.
evidenceRefs	Material assertions	Supporting, disputing and qualifying evidence.
confidenceState	Material assertions	Controlled confidence/UNKNOWN state.
reviewState	Proposed or approved content	Candidate, approved, rejected, superseded.
securityClassification	Sensitive objects and paths	Handling requirement.
supersedes	Versioned records	Historical lineage without overwrite.

Part V - Namespace, identifier and version strategy

16. Recommended namespaces

Prefix	Purpose	Illustrative namespace
aidcf:	AIDCF canonical ontology	https://w3id.org/aidcf/core/1.0/
aidcf-thf:	AIDCF threat/failure extension	https://w3id.org/aidcf/thf/1.0/
atg:	AI Trust Graph canonical ontology	https://w3id.org/aitrustgraph/core/1.0/
aix:	Integration bridge ontology	https://w3id.org/aidcf/integration/1.0/
prov:	W3C provenance vocabulary	http://www.w3.org/ns/prov#
skos:	Vocabulary mappings	http://www.w3.org/2004/02/skos/core#

The namespace locations above are proposed publication patterns and require maintainer approval and persistent-identifier configuration before release.

17. Identifier model

- Global IDs identify real-world or conceptual instances across views.
- Canonical term IDs remain in their source namespace.
- Mapping records receive independent IDs and lifecycle status.
- Path and run identifiers are immutable and versioned.
- Sensitive instance IDs should be opaque and must not reveal facility details.
- Retired identifiers are never reassigned.

18. Mapping semantics

Mapping type	Meaning	Permitted use
EXACT	Semantics equivalent for declared scope.	Requires joint maintainer approval.
CLOSE	Substantially aligned but differences remain.	Interoperability with disclosed limitation.

Mapping type	Meaning	Permitted use
BROADER	Source concept is broader than target.	Taxonomy navigation, not property inheritance.
NARROWER	Source concept is narrower than target.	Profile specialization.
RELATED	Material relationship without semantic equivalence.	Default for most cross-method links.
SHARED_INSTANCE	Two representations refer to the same real-world instance.	Identity federation with separate method views.
NO_MAP	Concept intentionally has no counterpart.	Boundary preservation.

INT-REQ-018 Mapped terms shall declare mapping type, source and target versions, rationale, approver and effective period.

INT-REQ-019 Close, broader, narrower and related mappings shall not drive automatic property equivalence.

INT-REQ-020 Breaking changes shall increment the major version and publish migration guidance.

Part VI - Node interoperability model

19. Node superclasses

Node superclass	Definition
aix:ContextNode	Assessment, scope, boundary, environment, decision and version context.
aix:CapabilityNode	Capability Instance, Service Outcome and Critical Function anchors.
aix:InfrastructureNode	Location, system, asset, component, interface and resource.
aix:DigitalSystemNode	AI system, model, agent, tool, data, prompt, pipeline, API and provider.
aix:ActorNode	Person, team, organisation, supplier, operator and assessor.
aix:IdentityNode	Human, workload, service, device and federated identities.
aix:AdverseConditionNode	Threat, hazard, cause, failure, effect and consequence.
aix:ProtectionNode	Safeguard, control, barrier, breakpoint and recovery action.
aix:AssuranceNode	Assertion, evidence, test, finding, decision and limitation.
aix:TemporalNode	Snapshot, change event, state and validity interval.
aix:IntegrationNode	Mapping, handoff, decision package, synchronization event and conflict.

20. Shared-entity identity rules

INT-REQ-021 A shared entity shall have one globalId and one or more method-specific views.

INT-REQ-022 Method views shall retain canonical class, owner, properties, evidence and lifecycle.

INT-REQ-023 A merged display shall be labelled integrated view and shall not replace native records.

INT-REQ-024 Conflicting properties shall remain separate assertions until resolved.

INT-REQ-025 Node granularity shall follow the decision question; aggregation shall preserve membership and consequence.

21. Primary shared-node candidates

Real-world subject	AIDCF view	ATG view	Integration rule
GPU compute node	Compute Asset	Workload/Infrastructure Target	Shared instance; distinct operational and authority properties.
Cluster manager	Compute Management System	Service/API/Privileged Target	Shared instance with control-plane interface.
BMS/EPMS/SCADA	Facility Control System	Service/Control Plane/Privileged Target	Shared instance; physical function remains AIDCF-owned.
Storage platform	Storage System	Data/Service Target	Shared instance; distinguish infrastructure availability from data access.
Network fabric controller	Network Management System	Control Plane/Privileged Target	Shared instance; distinguish fabric function from digital authority.
Operator	Human Dependency/Role	Human Actor/Identity	Shared person/role link; avoid replacing identity with job role.
Supplier	External Dependency	Provider/Organisation	Shared organisation with separate contractual and trust edges.

Real-world subject	AIDCF view	ATG view	Integration rule
Evidence item	Evidence Object	Evidence Object	Joint profile where lawful and relevant.
Assessment boundary	Infrastructure scope boundary	System/trust boundary	Related or overlapping boundaries, not automatically identical.

22. Node lifecycle states

State	Meaning
CANDIDATE	Proposed by collector, inference or analyst.
OBSERVED	Directly observed but not yet approved as canonical fact.
APPROVED	Reviewed and accepted for the declared scope/snapshot.
DISPUTED	Material conflict unresolved.
STALE	Evidence or state no longer sufficiently current.
SUPERSEDED	Replaced by a later approved version.
RETIRED	No longer active but retained historically.
UNKNOWN	Existence, class or material attribute unresolved.

Part VII - Relationship interoperability model

23. Edge families

Family	AIDCF examples	ATG examples	Bridge examples
Structural	PART_OF, LOCATED_IN, IMPLEMENTS	CONTAINS, DEPLOYED_IN, PART_OF	SHARED_INSTANCE_OF, METHOD_VIEW_OF.
Dependency	DEPENDS_ON, POWERED_BY, COOLED_BY	RELIES_ON, PROVIDED_BY	ENABLES, PROVIDES_INFRASTRUCTURE_SERVICE_TO.
Connectivity	CONNECTED_TO, INTERFACES_WITH	NETWORK_REACHES, CALLS, INVOKES	HAS_DIGITAL_INTERFACE, EXPOSES_ENDPOINT.
Identity/authority	OPERATED_BY, MAINTAINED_BY	AUTHENTICATES_AS, AUTHORIZES, DELEGATES_TO	AUTHORIZES_CONTROL_OF, OPERATES_THROUGH_IDENTITY.
Failure	CAUSES, PROPAGATES_TO, AFFECTS	ENABLES_PATH, AMPLIFIES, COMPROMISES	CREATES_DIGITAL_CONDITION, CAN_AFFECT.
Protection	PROTECTED_BY, CONTAINED_BY, RECOVERED_BY	CONTROLLED_BY, BLOCKED_BY, DETECTED_BY	REALIZED_BY, BREAKS_INTEGRATED_PATH.
Evidence	EVIDENCED_BY, CHALLENGED_BY	SUPPORTS, DISPUTES, QUALIFIES	SAME_EVIDENCE_OBJECT, RELEVANT_TO_VIEW.
Decision	RESULTS_IN, INVALIDATES	INFORMS, ACCEPTED_BY, SUPERSEDES	CONTRIBUTES_TO_DECISION_PACKAGE.

24. Bridge relationship registry

Bridge edge	Source	Target	Meaning
REPRESENTED_BY_AI_SYSTEM	CapabilityInstance	ATG SystemOfInterest	Links capability to digital system representation.
ENABLES_DIGITAL_FUNCTION	CriticalFunction/System	ATG Function/Object	Infrastructure function enables digital behavior.
PROVIDES_INFRASTRUCTURE_SERVICE_TO	AIDCF System	ATG Object	Power, cooling, compute, storage, network or facility service.
HAS_DIGITAL_INTERFACE	AIDCF Asset/System	ATG Service/API/Endpoint	Management or telemetry interface.
OPERATES_THROUGH_IDENTITY	AIDCF Actor/Role	ATG Identity	Identity used to exercise operational action.
AUTHORIZES_CONTROL_OF	ATG Identity/Authority	AIDCF Asset/System	Digital authority can command/configure infrastructure.
CREATES_DIGITAL_CONDITION	AIDCF Failure/Effect	ATG Condition	Infrastructure event changes digital state or trust conditions.
CAN_AFFECT	ATG Path/Actor/Tool	AIDCF Asset/Function/Capability	Digital path may cause physical or infrastructure consequence.
REALIZED_BY	ATG Control	AIDCF Safeguard	Control is implemented by real infrastructure measure.
BREAKS_INTEGRATED_PATH	Control/Safeguard	IntegratedPath	Verified interruption of cross-method progression.
METHOD_VIEW_OF	Method-specific node	Shared entity	Associates native view with global instance.
MAPS_TO	Canonical term	Canonical term	Controlled semantic mapping.
CONTRIBUTES_TO_DECISION_PACKAGE	Decision	DecisionPackage	Federates conclusions without aggregation.
TRIGGERS_REASSESSMENT_OF	Change/Finding	Assessment/Snapshot	Change invalidates or limits prior conclusion.

25. Edge minimum record

Field	Requirement
edgeId	Immutable identifier.
edgeType	Canonical native or bridge predicate.
source/target	Compatible endpoint IDs.
canonicalOwner	AIDCF, ATG or Integration.
direction	Explicit; inverse only if registered.
scope/conditions	Identity, state, protocol, environment, load, time or approval as relevant.
evidenceRefs	Supporting/disputing evidence.
confidenceState	Controlled confidence or UNKNOWN.
reviewState	Candidate through superseded.
effectivePeriod	Time for which assertion applies.
securityClassification	Handling requirement.
sourceVersion	Controlling artifact/schema version.

Part VIII - Boundary and interface model

26. Boundary classes

Boundary	AIDCF emphasis	ATG emphasis	Integrated question
Physical	Site, hall, cage, rack and route.	Physical access as trust precondition.	Can access create digital or physical consequence?
Electrical/thermal	Power and cooling fault/containment boundaries.	Dependency and consequence target.	Can failure cross into digital service or authority loss?
Network	Fabric, management and external connectivity.	Reachability, segmentation and protocol conditions.	Does connectivity plus identity create actionable control?
Identity	Operator accounts and management access dependencies.	Principal, credential, federation, delegation and privilege.	Who can act on infrastructure, under what conditions?
Administrative	Owner, operator, tenant and supplier responsibility.	Trust grant, provider and decision authority.	Where does accountability or enforcement change?
Provider	Utility, OEM, colocation and service provider.	AI/cloud/tool/data provider boundary.	Which external dependency can collapse multiple assumptions?
Sovereignty	Location, operation, maintenance and supply constraints.	Data flow, identity, provider and control jurisdiction.	Do physical and digital dependencies meet sovereignty intent?
Evidence	Facility data classification and disclosure limits.	Graph evidence, identity and finding sensitivity.	Can assurance remain traceable without exposing sensitive paths?

27. Boundary Crossing object

Property	Required content
crossingId	Stable identifier.
sourceBoundary/targetBoundary	Explicit boundary objects.
crossingEntity	Flow, identity, service, person, supplier, resource or path.
nativeRelation	AIDCF or ATG relationship.
bridgeRelation	Integration relation where applicable.
expectedEnforcement	Control, safeguard or authority gate.
operatingStates	Normal, degraded, maintenance, emergency, recovery or transition.
evidence	Design and observed evidence.
owner	Responsible and accountable parties.
failureModes	How crossing or enforcement may fail.
confidence/UNKNOWN	Current evidence-bounded state.

Part IX - Failure, dependency and consequence integration

28. Integrated scenario metamodel

Stage	Native owner	Required object
Decision context	Joint	DecisionQuestion, CapabilityInstance, ServiceOutcome, scope and snapshot.
Infrastructure context	AIDCF	Assets, systems, dependencies, fault domains, state and operating envelope.
Adverse initiation	AIDCF	Threat/hazard, initiating event, cause and failure mode.
Infrastructure propagation	AIDCF	Effects through explicit dependencies and fault domains.
Digital handoff	Integration	CREATES_DIGITAL_CONDITION or loss/change of interface, identity, telemetry or service.
Trust/authority propagation	ATG	Conditional path across identity, access, delegation, invocation, influence or trust.
Capability/business consequence	Joint	Technical, AI-system, service, stakeholder and strategic consequence.
Protection	Joint	Infrastructure safeguards, digital controls and cross-boundary breakpoints.
Evidence/decision	Joint	Assertions, evidence bundle, UNKNOWN, findings and separate/federated decisions.

29. Integrated path types

Path type	Description
Infrastructure-to-digital	Physical or operational failure alters digital service, identity, telemetry or control-plane condition.
Digital-to-infrastructure	Identity, tool or service authority can alter infrastructure configuration or operation.
Compound cyber-physical	Deliberate digital action combines with physical or engineering condition.
Common-dependency collapse	One utility, identity, controller, provider, team or service affects both methodologies.
Detection/attribution loss	Failure removes telemetry needed for AI Trust Graph evidence or infrastructure response.
Recovery-coupled	Digital recovery depends on physical infrastructure, or facility recovery depends on digital control/identity.
Sovereignty path	Physical location, operator, provider, support identity and data/control jurisdiction combine.
Change/transition path	Upgrade, expansion, migration or temporary bypass changes both infrastructure and trust state.

30. Cross-method causal discipline

INT-REQ-030 A native AIDCF causal relationship shall not be relabelled as an ATG authority relationship.

INT-REQ-031 An ATG reachability path shall not be treated as an infrastructure failure without an evidenced action and effect.

INT-REQ-032 Integrated scenarios shall identify the exact semantic handoff step.

INT-REQ-033 Every path step shall declare native or bridge semantics.

INT-REQ-034 Propagation may stop at UNKNOWN and shall not be completed by assumption.

INT-REQ-035 Controls and safeguards shall be evaluated against the specific path segment and operating state they cover.

31. Consequence alignment

AIDCF consequence	ATG consequence alignment	Integration note
Capability availability	System/service unavailability	Same event may have infrastructure and digital causal segments.
Compute capacity	Loss of AI processing/actionability	Do not equate capacity loss with compromise.
Data/artifact integrity	Data/model integrity consequence	Trace whether cause is storage failure, authority misuse or both.
Isolation/security	Boundary or trust collapse	Preserve physical and digital isolation semantics.
Recovery complexity	Containment/rollback/revocation difficulty	Model recovery dependencies across both methods.
Strategic/sovereignty	Provider, jurisdiction or authority exposure	Assess physical and digital sovereignty jointly.
Evidence/attribution loss	Observability and audit-path failure	Evidence degradation may cap both conclusions.

Part X - Trust, authority and infrastructure-control integration

32. Infrastructure authority model

Infrastructure authority is represented when a human or machine identity can observe, configure, command, isolate, bypass, update, stop, start, transfer, reset or otherwise affect an infrastructure asset or function. Authority remains distinct from network reachability and from role ownership.

Authority action	Examples of target	Required conditions
OBSERVE	BMS, EPMS, telemetry, logs	Identity, permission, interface, scope and evidence.
CONFIGURE	Setpoint, protection, route, firmware	Write authority, approval, change control and rollback.
COMMAND	Start/stop/transfer/isolate	Action permission, safety interlock and operating state.
BYPASS	Alarm, protection, redundancy, approval	Explicit authorization, duration, monitoring and restoration.
UPDATE	Firmware, image, policy, logic	Supply integrity, compatibility, approval and recovery.
REVOKE	Session, credential, remote support	Revocation mechanism and verified effect.
RECOVER	Restore configuration/service	Approved baseline, identity, sequence and validation.

33. Authority amplification patterns

Pattern	Integrated example	Required analysis
Shared management identity	One account controls multiple cooling, power or compute systems.	Blast radius, credential boundary, approval, telemetry and revocation.
Control-plane inheritance	Group or federated role inherits infrastructure write rights.	Delegation chain, scope, conditions and evidence.
Tool-mediated action	AI agent or automation invokes a management API.	Effective identity, tool scope, approval and reversibility.
Provider remote support	OEM identity can access privileged interfaces.	Contract, activation, monitoring, jurisdiction and revocation.
Emergency bypass	Operational role gains exceptional authority.	Trigger, duration, independent review and cleanup.
Telemetry-to-command coupling	Monitoring platform can issue automated control actions.	Sensor trust, logic, fail state and common dependency.

34. Control and safeguard alignment

AIDCF safeguard function	ATG control function	Integrated use
Anticipate	Govern/classify	Identify preconditions and high-consequence paths.
Prevent	Block/constrain	Stop failure or unauthorized action before progression.
Detect	Observe/alert	Detect physical, digital or integrated state change.
Decide	Approve/authorize	Apply human or machine decision gate.
Respond	Revoke/isolate	Constrain active progression.
Contain	Segment/limit blast radius	Keep effect within fault/trust boundary.
Withstand	Maintain service/least authority	Sustain minimum outcome while limiting actionability.
Recover	Restore/rollback	Return infrastructure and digital system to approved state.
Adapt	Relearn/reconfigure	Update models, controls, architecture and evidence after change.

Part XI - Evidence, provenance, confidence and UNKNOWN

35. Common evidence profile

Field family	Minimum content
Identity	evidenceId, type, title and canonical owner.
Source	Source system, owner, authority and original reference.
Collection	Collector, method, authorization, date and query/procedure.
Scope	Assets, identities, environment, population, period and states.
Integrity	Hash/signature, custody and transformation history.

Field family	Minimum content
Quality	Grade, relevance, currentness, corroboration and representativeness.
Protection	Classification, privacy, need-to-know, retention and legal hold.
Traceability	Assertions, nodes, edges, paths, controls, findings and decisions supported/disputed.
Review	Reviewer, outcome, conflicts, limitations and supersession.

36. Provenance alignment

The integration profile should use a provenance model capable of distinguishing entities, activities and responsible agents. Where RDF/OWL is used, W3C PROV-O is the preferred external vocabulary for interoperable provenance links, with AIDCF/ATG-specific specialisations where needed.

37. Evidence grades and state mapping

Grade/state	Integration interpretation
E0	No usable evidence. Relevant assertion remains UNKNOWN or Not Tested.
E1	Inference or uncorroborated signal. Candidate content only.
E2	Accountable attestation. Supports claimed practice, not independent effectiveness.
E3	Approved documentation. Supports design/governance intent.
E4	Corroborated technical evidence. May support implementation within observed scope.
E5	Direct current technical evidence plus representative test/operation. May support bounded effectiveness.
CONFLICTING	Material sources disagree; do not average or pick silently.
STALE	Evidence no longer sufficiently representative after time or change.
UNKNOWN	Insufficient evidence or unresolved conflict; preserved end to end.

38. Assertion contract

Property	Requirement
assertionId	Stable versioned identifier.
subject/predicate/object	One precise proposition using canonical terms.
methodOwner	AIDCF, ATG or Integration.
scope/conditions	Boundary, environment, state, identity, configuration and time.
evidenceRefs	Supporting, disputing and qualifying evidence.
confidence	Controlled state plus rationale.
review	Candidate, approved, rejected, modified or superseded.
limitations	Explicit non-covered conditions.
derivedFrom	Transformation/query/rule and source version.
decisionImpact	Which finding, path or decision relies upon it.

INT-REQ-038 Transformation shall preserve source IDs, provenance, evidence grade, confidence and review state.

INT-REQ-039 AI-generated extraction or inference shall create a candidate assertion and shall not overwrite approved facts.

INT-REQ-040 One evidence item may support both methods only after assertion-specific relevance and scope are evaluated.

INT-REQ-041 Conflicting evidence shall remain queryable after resolution.

INT-REQ-042 Evidence classification may restrict graph display without deleting traceability.

Part XII - Path, scenario and query semantics

39. IntegratedPath object

Field	Required content
pathId	Immutable ID and version.
pathType	Infrastructure-to-digital, digital-to-infrastructure, compound, etc.
startCondition	Actor, threat, hazard, failure, identity or state.
orderedSteps	Native nodes/edges and bridge handoff steps.
target	Capability, asset, privileged action, data, stakeholder or service outcome.
preconditions	Identity, authority, protocol, load, state, environment and timing.
boundariesCrossed	Physical, network, identity, provider, administrative or sovereignty.
breakpoints	Controls/safeguards attached to specific steps.
evidenceCoverage	Evidence per material step.
unknownSteps	Unverified elements and impact.
residualPath	Remaining progression after controls.
reviewState	Candidate, validated, disputed, superseded.
snapshot	Graph/version context.

40. Path validity rules

PATH-001 Every adjacent step shall be type-compatible.

PATH-002 Every bridge step shall use an approved integration relationship.

PATH-003 A topological connection shall not imply executable authority.

PATH-004 A reachable digital target shall not imply realised infrastructure effect.

PATH-005 Conditions and operating states shall be evaluated at each material step.

PATH-006 A control shall interrupt only the path conditions it has demonstrably covered.

PATH-007 Alternative and common-dependency paths shall be considered before closure.

PATH-008 A path with critical UNKNOWN may be prioritised but not declared proven.

PATH-009 Historical paths shall remain bound to their graph snapshot.

PATH-010 Sensitive paths shall be access controlled and safely abstracted for broad reports.

41. Canonical query families

Query family	Decision question
Capability dependency	Which infrastructure and external dependencies support this AI capability?
Physical-to-digital failure	Which digital services, identities or AI components are affected by loss of this infrastructure function?

Query family	Decision question
Digital-to-physical authority	Which identities, agents, tools or providers can affect critical facility or compute functions?
Common dependency	Which safeguards and controls share a utility, identity, controller, provider, team or management service?
Control breakpoint	Which verified intervention interrupts the most material integrated paths?
Critical UNKNOWN	Which unverified bridge relationships cap assurance?
Recovery chain	Which identities, systems, spares, services and infrastructure conditions are required to restore capability?
Evidence coverage	Which nodes, edges and path steps lack current evidence?
Graph drift	Which changed entities or relationships invalidate a prior integrated decision?
Sovereignty	Which cross-border physical, support, identity, provider or data dependencies affect the stated sovereignty claim?

42. Query result contract

- Query ID and version.
- Decision question and authorized purpose.
- Graph snapshot and schema versions.
- Start/target scope and filters.
- Returned nodes, edges and paths with native ownership.
- Evidence and confidence per material assertion.
- UNKNOWN and excluded data.
- Security classification.
- Reviewer and disposition where used for a finding or decision.

Part XIII - Lifecycle, synchronization and graph drift

43. Snapshot model

Snapshot	Meaning
AIDCF snapshot	Infrastructure configuration, state, evidence and scenario view at a point or period.
ATG snapshot	AI ecosystem objects, trust/authority edges, evidence and path view.
Integrated snapshot	References compatible native snapshots plus bridge assertions; does not copy them into a new hidden authority.
Decision snapshot	Frozen integrated subgraph used for a named decision.
Comparison snapshot	Pair/set of snapshots used for drift and reassessment analysis.

44. Synchronization modes

Mode	Use	Constraint
Manual controlled import	Review-led exchange of approved packages.	Full provenance and validation required.
Scheduled synchronization	Periodic exchange of changed approved records.	No implication of continuous monitoring.

Mode	Use	Constraint
Event-driven update	Change event proposes affected bridge assertions.	Human review for material facts/decisions.
Federated query	Query native repositories without copying all data.	Access, performance and snapshot consistency declared.
Offline evidence package	Sanitised package for independent assessment.	Immutable manifest, classification and completeness limitations.

45. Drift taxonomy

Drift	Example	Required response
Entity drift	New controller, identity, rack or provider.	Discovery and mapping review.
Relationship drift	Changed permission, route, dependency or operator.	Path re-analysis.
State drift	Maintenance, bypass, degraded cooling or emergency access.	State-specific re-evaluation.
Evidence drift	Evidence stale, superseded or conflicting.	Confidence review.
Control drift	Control disabled, scope reduced, exception added.	Breakpoint and residual-path review.
Schema drift	Class, edge or property version changes.	Compatibility and migration review.
Boundary drift	Ownership, tenant, network, provider or jurisdiction changes.	Boundary-crossing review.
Capability drift	Workload, scale, autonomy or service outcome changes.	Rebaseline AIDCF and ATG scopes.

46. Reassessment triggers

- Material AIDCF Change Event.
- New or changed ATG identity, authority, tool, agent or provider relationship.
- Critical incident or near miss.
- Failure of a safeguard or control breakpoint.
- Evidence expiry or conflict.
- Capacity expansion or new liquid-cooled/high-density deployment.
- Provider, jurisdiction or remote-support change.
- Schema major version or mapping change.
- Resolution of a critical UNKNOWN.

Part XIV - Data exchange, API and machine-readable package

47. Required publication package

Artifact	Purpose
aidcf-atg-integration-v1.0.owl	Bridge ontology and formal class/property declarations.
aidcf-atg-integration-v1.0.ttl	Readable RDF/Turtle serialization.
aidcf-atg-context-v1.0.jsonld	JSON-LD context.
aidcf-atg-exchange-v1.0.schema.json	JSON exchange contract.
aidcf-atg-shapes-v1.0.ttl	SHACL validation shapes.

Artifact	Purpose
class-mappings-v1.0.csv	Class mapping registry.
relationship-mappings-v1.0.csv	Relationship and bridge edge registry.
query-catalog-v1.0.yaml	Named query definitions and expected outputs.
conformance-tests-v1.0/	Positive, negative and migration fixtures.
examples/	Synthetic integrated profiles and decision packages.
CHANGELOG.md	Versioned change history.
MIGRATION.md	Compatibility and migration instructions.
SECURITY.md	Sensitive-path and vulnerability disclosure guidance.
LICENSE.md	Rights and obligations for public reuse.

48. Exchange envelope

```
{
  "packageId": "aix-package:example-001",
  "schemaVersion": "1.0",
  "aidcfSnapshot": "aidcf-snapshot:001",
  "atgSnapshot": "atg-snapshot:014",
  "bridgeAssertions": ["aix-edge:001"],
  "evidenceManifest": ["evidence:001"],
  "unknowns": ["unknown:001"],
  "securityClassification": "RESTRICTED",
  "reviewState": "CANDIDATE"
}
```

49. API resource model

Resource	Minimum operations / behavior
/mappings	Retrieve approved mappings by version and status.
/shared-entities	Resolve global ID to native method views.
/bridge-assertions	Create candidate, review, approve, supersede and query bridge assertions.
/snapshots	Register immutable native and integrated snapshots.
/paths	Submit/query integrated paths and validation state.
/evidence-manifests	Reference evidence metadata without exposing protected content.
/unknowns	Track unresolved information and assurance impact.
/decision-packages	Assemble separate conclusions and shared limitations.
/conformance	Validate package/schema/mapping/test profile.

This resource model is normative at the information-contract level only. It does not prescribe transport protocol, authentication product, database or deployment topology.

50. OWL, SHACL and JSON roles

Technology	Role	Anti-error rule
OWL/RDF	Semantic classes, properties, mappings and reasoning.	Open-world reasoning does not validate required record completeness.
SHACL	Validate RDF data shapes, cardinality and compatibility.	Validation shapes do not prove real-world truth or control effectiveness.
JSON Schema	Validate exchange structure and datatypes.	Structural validity is not semantic or assurance validity.
JSON-LD	Carry linked identifiers in JSON exchange.	Context resolution and version must be controlled.
PROV-O	Represent interoperable provenance entities, activities and agents.	Specialization shall preserve provenance semantics.

Part XV - Security, privacy and sensitive-path protection

51. Information sensitivity

Information class	Examples	Minimum handling
Public methodology	Canonical terms, synthetic examples and abstract patterns.	Approved public repository.
Controlled methodology	Draft mappings, unresolved decisions and review notes.	Maintainer access.
Restricted architecture	Facility layout, power/cooling paths, identities, routes and providers.	Need-to-know, encryption and monitored access.
Highly restricted path	Combined physical and digital path to critical control or capability.	Segregated access, strong approval and limited export.
Evidence protected	Logs, configs, contracts, telemetry and personal data.	Lawful purpose, minimisation, retention and access controls.
Security vulnerability	Actionable weakness or live exploit path.	Coordinated disclosure and restricted remediation workflow.

52. Security requirements

SEC-001 Integrated-path access shall be authorised by consequence and sensitivity, not only by repository membership.

SEC-002 Exports shall support redaction or abstraction while retaining reference integrity.

SEC-003 Credentials, secrets and sensitive values shall not be stored as ordinary node properties.

SEC-004 Query and export activity for restricted graphs shall be auditable.

SEC-005 Tenant and client data shall remain logically and cryptographically separated where required.

SEC-006 Synthetic examples shall not reveal identifiable facility or client details.

SEC-007 Inference and automation shall operate within explicit data-handling and output-release controls.

SEC-008 Deletion and retention actions shall preserve required audit lineage and legal obligations.

53. Safe publication pattern

- Publish canonical schemas, mappings and synthetic examples.
- Do not publish instance graphs of real facilities or live privilege paths.
- Separate public identifiers from operational identifiers.
- Provide abstract path narratives where detailed disclosure creates risk.
- Perform employer, client, IP, privacy, security and licensing review before release.

Part XVI - Conformance, validation and test suite

54. Core validation catalogue

Rule	Validation statement
VAL-001	Every mapping declares source and target artifact versions.
VAL-002	Every shared instance has one globalId and at least one native method view.
VAL-003	Every bridge edge has allowed source and target classes.

Rule	Validation statement
VAL-004	Every material bridge assertion has evidence, confidence, review and effective period.
VAL-005	Every integrated path contains at least one explicit approved handoff edge.
VAL-006	Every path step identifies native owner or Integration ownership.
VAL-007	No digital reachability edge is treated as physical effect without action/effect evidence.
VAL-008	No physical connection is treated as digital authority without identity/permission evidence.
VAL-009	Every control-break claim identifies covered path segment and test conditions.
VAL-010	Every unresolved critical step is represented as UNKNOWN.
VAL-011	Every integrated decision package retains separate native conclusions.
VAL-012	Every snapshot references immutable native graph versions.
VAL-013	Every supersession retains prior record and rationale.
VAL-014	Every restricted path carries security classification.
VAL-015	Every AI-generated assertion remains candidate until approved.
VAL-016	Every evidence reuse is reviewed for assertion-specific relevance.
VAL-017	Every schema major-version change has migration guidance.
VAL-018	Every mapping conflict is recorded as OPEN, disputed or resolved.
VAL-019	Every boundary crossing identifies expected enforcement and owner.
VAL-020	Every synchronization run produces audit and validation results.

55. Positive conformance tests

- Resolve a shared compute manager into AIDCF and ATG views.
- Build an infrastructure-to-digital failure path with one bridge handoff.
- Build a digital-to-infrastructure authority path with evidenced identity and permission.
- Attach one evidence item to separate assertions with different scope ratings.
- Supersede a bridge assertion without deleting history.
- Compare integrated snapshots and identify drift-triggered reassessment.
- Produce a decision package with separate AIDCF, ATG and integrated limitations.

56. Negative tests

- Reject bridge edge with incompatible endpoints.
- Reject approved material assertion lacking evidence or explicit UNKNOWN.
- Reject path that jumps from network connection to control authority.
- Reject path closure based solely on policy evidence.
- Reject mapping that silently upgrades RELATED to EXACT.
- Reject package that replaces UNKNOWN with null, false or zero.
- Reject snapshot mutation after decision finalisation.
- Reject export of restricted path without authorization/classification.

57. Conformance report

Field	Required content
implementationId/version	Named implementation and release.
profile/level	I0-I5 and optional profile.
schema versions	AIDCF, ATG and integration versions.
tests executed	Positive, negative, security and migration tests.
results	Pass, fail, not tested and UNKNOWN.
exceptions	Approved deviations and rationale.
evidence	Logs, fixtures, reviewer and date.
limitations	Unsupported features and interoperability impact.
approval	Technical and methodology reviewers.
expiry/retest	Triggers and validity conditions.

Part XVII - Governance, change and release management

58. Governance bodies and roles

Role/body	Accountability
AIDCF methodology maintainer	Approves AIDCF meanings and native mappings.
AI Trust Graph methodology maintainer	Approves ATG meanings and native mappings.
Integration architecture board	Owns bridge ontology, mapping conflicts and conformance profile.
Domain SMEs	Review engineering, compute, storage, networking, OT, identity and AI semantics.
Evidence and assurance reviewer	Reviews provenance, confidence, UNKNOWN and decision discipline.
Security reviewer	Reviews sensitive path, access, export and disclosure controls.
Legal/IP/publication reviewer	Reviews ownership, licensing, employer/client and public claims.
Release manager	Produces versioned package, changelog, signatures and migration artifacts.

59. Change classes

Class	Example	Version impact
Editorial	Clarification without semantic change.	Patch.
Additive compatible	New bridge mapping or optional property.	Minor.
Constraining compatible	New validation shape that formalises existing requirement.	Minor with impact note.
Breaking semantic	Changed meaning, endpoint compatibility or required property.	Major.
Security emergency	Restriction or deprecation to prevent harmful exposure.	Patch/minor plus urgent advisory.

Class	Example	Version impact
Deprecation	Term remains readable but replacement specified.	Minor; removal only in major.

60. Mapping proposal record

Field	Content
proposalId	Stable ID.
source/target	Canonical term IDs and versions.
mappingType	EXACT, CLOSE, BROADER, NARROWER, RELATED or NO_MAP.
rationale	Decision use and semantic comparison.
differences	Scope, properties, conditions and exclusions.
impact	Queries, profiles, data migration and decisions affected.
evidence	Examples and validation tests.
reviewers	AIDCF, ATG, ontology and domain reviewers.
status	Proposed, approved, rejected, deprecated or superseded.
effectiveVersion	Integration release in which valid.

61. Release gates

- Semantic consistency with all six canonical dependencies.
- No unresolved breaking conflicts hidden in mappings.
- Schema, shapes, mappings and examples pass conformance tests.
- Synthetic examples reviewed for safety and validity.
- Public/private boundary verified.
- Employer, IP, confidentiality, privacy, legal, trademark and license review completed.
- Migration and changelog published.
- Independent practitioner usability review completed.

Part XVIII - Reference profiles and worked examples

62. Reference profiles

Profile	Focus	Mandatory bridge objects
AI factory training	Distributed compute, parallel storage, high-speed fabric, power/cooling and scheduler.	Capability, infrastructure services, management identities, failure-to-digital handoff, checkpoint consequence.
AI colocation	Provider/tenant responsibilities and evidence access.	Provider boundaries, shared instances, contracts, remote identities and UNKNOWN ownership.
Sovereign AI	Location, operator control, provider support and jurisdiction.	Sovereignty boundaries, external dependencies, identities, data/control crossings and decision package.
Agentic operations	Agent/tool authority affecting infrastructure or capacity.	Agent, tool, identity, approval, target asset, action, safeguard/control and rollback.
Facility OT management	BMS/EPMS/SCADA digital control of engineering systems.	Control interface, identities, authority, failure modes, telemetry and containment.
Recovery assurance	Restore AI capability after compound disruption.	Recovery dependencies, identities, golden configs, spares, infrastructure state and evidence.

63. Worked example A: cooling degradation to AI consequence

Step	Native semantics	Example assertion
1	AIDCF State	Cooling system is DEGRADED under sustained training load.
2	AIDCF Failure	CDU capacity loss creates thermal effect on rack-scale system.
3	Integration handoff	Thermal effect CREATES_DIGITAL_CONDITION: accelerator throttling/health alarms.
4	ATG System path	Scheduler and observability services receive reduced/unstable capacity signals.
5	ATG Authority/decision	Automation may reschedule, throttle or terminate jobs under approved policy.
6	Joint consequence	Required training completion and checkpoint outcomes are at risk.
7	Protection	AIDCF safeguard and ATG approval/control breakpoint constrain progression.
8	Evidence	Facility telemetry, GPU telemetry, scheduler events and test record support separate assertions.
9	Decision	AIDCF and ATG conclusions remain separate; integrated decision package identifies combined limitation.

64. Worked example B: privileged remote support to physical effect

Step	Native semantics	Example assertion
1	ATG Actor/Identity	OEM support identity is activated for a defined maintenance window.
2	ATG Reachability	Identity can authenticate to management gateway and reach a controller API.
3	ATG Authority	Permission permits configuration change to the cooling control system.
4	Integration handoff	Authority AUTHORIZES_CONTROL_OF an AIDCF cooling controller.
5	AIDCF Effect	Incorrect setpoint can degrade heat removal under high load.
6	AIDCF Propagation	Thermal effect propagates to compute capacity and service outcome.
7	Controls	Just-in-time activation, approval, command allowlist, independent telemetry and revocation.
8	Evidence	Identity logs, permission export, change record, controller logs and thermal telemetry.
9	UNKNOWN	If command scope is not evidenced, path remains candidate and assurance is capped.

65. Worked example C: evidence conflict

Source	Assertion	Treatment
Approved architecture	Management network is isolated from enterprise access.	E3 supports design intent.
Observed route export	A restricted gateway route exists.	E4 challenges current isolation assertion.
Identity inventory	Accessible identity scope is incomplete.	UNKNOWN for who can use route.
Integration outcome	Retain design and observed assertions separately; mark conflict and create candidate path.	
Decision	Do not declare exploitable access or effective isolation until identity/permission and tested conditions are resolved.	

Appendix A - Class mapping registry

AIDCF class	ATG class	Mapping	Rule
aidcf:CapabilityInstance	atg:SystemOfInterest/AISystem	CLOSE/RELATED	Shared capability anchor; not universal equivalence.
aidcf:ServiceOutcome	atg:Outcome	CLOSE	Outcome semantics aligned with method-specific properties.
aidcf:CriticalFunction	atg:Function	CLOSE	AIDCF function is infrastructure/capability centred.
aidcf:AssetInstance	atg:Asset/Object	BROADER/NARROWER	Use shared instance mapping for real systems.

AIDCF class	ATG class	Mapping	Rule
aidcf:InfrastructureSystem	atg:InfrastructureObject	CLOSE	ATG view adds trust/authority semantics.
aidcf:Dependency	atg:Dependency	CLOSE	Do not treat as TrustGrant.
aidcf:AssessmentBoundary	atg:Boundary	RELATED	Boundary dimensions may overlap but differ.
aidcf:OperatingState	atg:Condition/State	CLOSE	Preserve AIDCF enumerations.
aidcf-thf:Threat	atg:ThreatCandidate	RELATED	Threat analysis scope differs.
aidcf-thf:FailureMode	atg:FailureCondition	CLOSE	Use AIDCF canonical failure distinctions.
aidcf-thf:PropagationPath	atg:Path	RELATED	Failure path differs from trust/exposure path.
aidcf:Safeguard	atg:Control/ControlBreakpoint	RELATED	Real measure vs assessment/control abstraction.
aidcf:EvidenceObject	atg:EvidenceItem	CLOSE	Joint evidence profile may be used.
aidcf:Finding	atg:Finding	CLOSE	Keep source method and decision basis.
aidcf:AssuranceDecision	atg:Decision	CLOSE	Federate, do not aggregate automatically.
aidcf:Unknown	atg:UNKNOWN	EXACT within shared profile	Preserve as first-class state.

Appendix B - Relationship mapping registry

AIDCF/bridge	ATG	Mapping	Constraint
aidcf:DEPENDS_ON	atg:RELIES_ON	CLOSE	Preserve dependency type and strength.
aidcf:CONNECTED_TO	atg:NETWORK_REACHES	RELATED	Connection does not prove network reachability.
aidcf:INTERFACES_WITH	atg:CALLS/INVOKES	RELATED	Interface existence does not prove invocation.
aidcf:OPERATED_BY	atg:ACTS_AS/USES_IDENTITY	RELATED	Operation role does not define effective identity.
aidcf:PROTECTED_BY	atg:CONTROLLED_BY	RELATED	Safeguard existence does not prove control effectiveness.
aidcf:EVIDENCED_BY	atg:SUPPORTS	RELATED	Evidence supports a precise assertion, not an object in general.
aidcf-thf:PROPAGATES_TO	atg:ENABLES_PATH	RELATED	Failure propagation differs from trust-path enablement.
aidcf-thf:AFFECTS	atg:IMPACTS	CLOSE	Scope and consequence must be stated.
aix:AUTHORIZES_CONTROL_OF	No native equivalent	BRIDGE	Digital authority targets infrastructure control.
aix:CREATES_DIGITAL_CONDITION	No native equivalent	BRIDGE	Infrastructure effect alters digital condition.
aix:REALIZED_BY	No native equivalent	BRIDGE	Digital control is realised by infrastructure safeguard.
aix:METHOD_VIEW_OF	No native equivalent	BRIDGE	Method-specific view references shared entity.

Appendix C - Minimum record schemas

C.1 Mapping record

Property	Cardinality	Type
mappingId	1	Identifier
sourceTerm	1	IRI
targetTerm	1	IRI
mappingType	1	Enumeration
sourceVersion	1	Version
targetVersion	1	Version
rationale	1	String
differences	0..*	String
reviewers	1..*	Party
status	1	Lifecycle state
effectivePeriod	1	Interval
tests	1..*	Test reference

C.2 Bridge assertion

Property	Cardinality	Type
bridgeAssertionId	1	Identifier
subject	1	Native/shared entity
predicate	1	Approved bridge edge
object	1	Native/shared entity
scope	1	Scope
conditions	0..*	Condition
evidence	1..* or UNKNOWN	Evidence reference
confidence	1	State
reviewState	1	State
effectivePeriod	1	Interval
securityClassification	1	Enumeration
sourceSnapshot	1..2	Snapshot references

C.3 Integrated decision package

Property	Cardinality	Type
packageId	1	Identifier
decisionQuestion	1	String/reference
aidcfDecision	0..1	Native Decision
atgDecision	0..1	Native Decision
integratedFindings	0..*	Finding
criticalUnknowns	0..*	UNKNOWN
evidenceManifest	1	Manifest
limitations	1..*	String
decisionOwner	1	Party
reviewers	1..*	Party
snapshot	1	Integrated Snapshot
status	1	State

Appendix D - Query pattern library

ID	Purpose
Q-001 Critical infrastructure authority	Find ATG identities/actors with AUTHORIZES_CONTROL_OF paths to AIDCF critical assets.
Q-002 Failure-to-AI impact	Traverse AIDCF PROPAGATES_TO then CREATES_DIGITAL_CONDITION to affected ATG systems and service outcomes.
Q-003 Shared control dependency	Find AIDCF safeguards and ATG controls that DEPEND_ON one identity, controller, provider or telemetry service.
Q-004 Missing bridge evidence	Find approved/candidate bridge assertions with E0/E1, stale, conflicting or UNKNOWN support.
Q-005 Boundary enforcement gap	Find crossings without expected enforcement, owner or current evidence.
Q-006 Recovery dependency chain	Find RECOVERED_BY and ATG authority/service paths required to restore a Capability Instance.
Q-007 Integrated single point	Find entities whose loss or compromise disconnects the capability or defeats multiple controls.
Q-008 Drift invalidation	Compare snapshots and return changed bridge edges used by final decisions.
Q-009 Residual path	Find alternate routes after a control or safeguard is applied.
Q-010 Sovereignty dependency	Find cross-jurisdiction provider, identity, support, data or infrastructure dependencies for a capability.

Appendix E - SHACL-style validation catalogue

Shape	Target	Constraint
SharedEntityShape	aix:SharedEntity	Exactly one globalId; at least one method view; canonical owner declared.
BridgeEdgeShape	aix:BridgeAssertion	Approved predicate; compatible endpoints; scope, evidence/UNKNOWN, confidence, review and time.
IntegratedPathShape	aix:IntegratedPath	Start, target, ordered steps, at least one bridge edge, snapshot and evidence coverage.
DecisionPackageShape	aix:DecisionPackage	Decision question, snapshot, evidence manifest, limitations and owner.
MappingShape	aix:Mapping	Source/target versions, mapping type, rationale, reviewer, status and tests.
UnknownShape	aix:Unknown	Unresolved attribute, materiality, owner if known and assurance impact.
SensitivePathShape	aix:IntegratedPath	Classification and access/export policy when target or steps are restricted.
SnapshotShape	aix:IntegratedSnapshot	Immutable IDs for compatible AIDCF and ATG snapshots.
EvidenceShape	aix:EvidenceObject	Source, collection, scope, integrity, grade, protection, review and assertion links.
ChangeShape	aix:SynchronizationEvent	Changed records, validation result, actor, time, source versions and audit reference.

Appendix F - Review and acceptance record

Review role	Status	Focus
Methodology author	Draft completed	Overall coherence.
AIDCF ontology reviewer	Pending	AIDCF term and relationship integrity.
AIDCF failure-model reviewer	Pending	Scenario and propagation semantics.
AI Trust Graph ontology reviewer	Pending	ATG object, edge and path integrity.
Data-centre engineering reviewer	Pending	Infrastructure handoff accuracy.
AI/security architecture reviewer	Pending	Identity, authority, trust and reachability.
OT/control-system security reviewer	Pending	BMS/EPMS/SCADA bridge risk.
Evidence/assurance reviewer	Pending	Provenance, confidence and decisions.
Knowledge-graph engineer	Pending	OWL/RDF/JSON-LD/SHACL implementation.
Security/privacy reviewer	Pending	Sensitive paths and data handling.
Legal/IP/publication reviewer	Pending	Rights, licensing and claims.
Independent practitioner review	Pending	Usability and interoperability.
Public release	Not approved	All mandatory gates closed.

Acceptance criteria

- Artifact 005 preserves the separate authority and independent usability of both methodologies.
- All shared concepts have a canonical owner and mapping type.
- Cross-method paths require explicit handoff edges and evidence.
- Physical connectivity, digital reachability, authority and realised consequence remain distinct.
- Evidence, confidence, UNKNOWN and supersession survive exchange.
- Machine-readable artifacts can be generated from the registries without inventing new semantics.
- Synthetic examples pass positive and negative conformance tests.
- Sensitive-path protection and publication boundaries are explicit.
- Multidisciplinary, legal, IP and security reviews are recorded before release.

Change record

Version	Status	Summary
0.1	Architecture draft	Established method boundaries, integration patterns and shared anchors.
0.5	Integrated draft	Added mappings, bridge relationships, path semantics, validation and profiles.
1.0	Controlled-review draft	Comprehensive normative baseline for expert review and machine-readable implementation.

Final doctrine

AIDCF and AI Trust Graph meet at the boundary between infrastructure capability and digital trust. Their integration must reveal how physical systems, operational dependencies, identities, digital authority, failure, exposure, controls and evidence combine to create or constrain consequence. Integration does not mean semantic collapse. The correct result is a traceable federation of native facts, explicit bridge assertions, evidence-bounded paths, preserved uncertainty and accountable decisions.